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Cointegration Capacity Utilization Paper Transparencies

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COINTEGRATION
Capacity Utilization Paper:
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*A General Method for Measuring Capacity Utilization: Application to the
OECD countries*

**~~Measuring Capacity Utilization in OECD Countries: A Cointegration
Method~~**

10/7/2004

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Measuring Capacity Utilization in OECD Countries: A Cointegration Method

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Abstract

This paper derives measures of potential output and capacity utilization for a number of OECD countries, using a method based on the cointegration relation between output and the capital stock. The intuitive idea is that economic capacity (potential output) is the aspect of output that co-varies with the capital stock over the long run. We show that this notion can be derived from a simple model that allows for a changing capital-capacity ratio in response to partially exogenous, partially embodied, technical change. Our method provides a simple and general procedure for estimating capacity utilization. It also closely replicates a previously developed census-based measure of U.S. manufacturing capacity-utilization. Of particular interest is that our measures of capacity utilization are very different from those based on aggregate production functions, such as the ones provided by the IMF.

It allows us to distinguish between classical, PK & Harrodian models of LR eq.

JEL Codes: D24, C52, C22, O57, O30

I. Introduction

Growth and technological change go hand-in-hand, because it is new investment that generally ushers in new technologies. But capital accumulation is inherently turbulent, subject to a variety of intrinsic cycles (van Duijn, 1983, ch 1) as well as conjunctural events such as wars, economic policies, and natural events. Therefore, in order to identify the slower and steadier influence of structural change, one must somehow adjust for the cyclical and conjunctural fluctuations. Since the latter variations are generally expressed as fluctuations in capacity utilization, one way to distinguish between short run influences and structural ones is to develop measures of capacity utilization and of economic capacity. Such considerations are critical to all long run analysis.

This paper aims to develop a simple general methodology for measuring capacity. The essential idea is that 'capacity' is the aspect of output that co-varies with the capital stock over the long run. This makes cointegration a natural tool for measuring capacity. ^{LR} ^{economic} ^(find better name) ^{We use the techniques} ^{IT MMS} ^{developed by PSS ...} ^{here was} previously developed for US data, and ^{proved} to provide capacity utilization measures that are remarkably close to completely independent measures based on the rate of utilization of electric motors used to drive capital equipment in manufacturing and on unpublished survey data from the McGraw-Hill (Shaikh 1987, 1992, 1999). In this paper we extend the results to encompass various OECD countries. ^{Shaikh..} ^{probably} ^{use..}

It is important at the outset to distinguish between 'engineering capacity', which is the maximum sustained production possible over some interval, and 'economic capacity', which is the *desired* level of output from given plant and equipment. For instance, it may be physically feasible to operate a plant for 20 hours per day 6 days a week, for a total of 120-hrs per week of engineering capacity. But it may turn out that the potentially higher costs of second and third shifts make it most profitable to operate only a single 8-hr shift per day for five days a week, i.e.

40-hrs per week. This is what defines economic capacity, the firm's benchmark level of output. Equivalently, economic capacity would be represented by an engineering capacity utilization rate of 33.33 % (40/120). Production persistently below this level would signal the need for a slowdown in planned capacity growth¹, while that persistently above it would signal the need for accelerated capacity growth (Foss 1963, p. 25; Shapiro 1989, p. 184; Kurz 1986, pp. 37-38, 43-44).

It should also be noted that economic capacity is not the same thing as "full employment output"². Since both measures have been labeled 'potential output', it is important to distinguish them. Even though the full capacity and full employment are common assumptions of standard economic theory, in actual practice there is no reason to suppose that production at economic capacity would serve to fully employ the existing labor force. Indeed, within classical, Keynesian, and Kaleckian traditions, the two are not synonymous even at the theoretical level (Garegnani, 1979; Shaikh, 1988).

It might be supposed that one could distinguish between actual output and capacity by means of some appropriate filtering technique. Indeed, if all actual fluctuations were roughly symmetric, then this might work. But actual data may not only contain multiple cycles, some as long as 20 years, but also generally exhibits a long term trend (which itself may or may not be part of some longer "wave"). Most techniques for identifying cycles therefore require that the data first be detrended. One common approach is to define the trend as some *a priori* function of time. But there is little reason to believe that growth trends are independent of actual rates of

¹ In a growing system, adjustments take place by changes in relative *growth* rates.

² Short run fluctuations in employment are likely to be correlated with short run fluctuations in output. Thus short run fluctuations in the employment rate (employment over labor supply) are likely to be correlated with short run fluctuation in the capacity utilization rate (output over capacity). But unless the ratio of capacity to labor supply, which is a kind of potential productivity of labor, happened to be always constant over time, we can be sure that the capacity utilization rate would deviate from the employment rate over the medium and long runs.

growth, and there is no real reason to prefer one time function over any other. Another procedure is to smooth the data so as to bring out the trend. But here we face the difficulty of disentangling the unknown trend from a multiplicity of cycles, ranging from 3-5 year inventory cycles to 20 year fixed capital cycles. Taking out the most rapid fluctuations leaves in all the longer ones mixed together with the trend. On the other hand, focusing on a particular longer cycle leaves in all of the shorter ones. Moreover, it is well known that various smoothing methods can give rise to spurious long cycles. Finally, there is the problem that fluctuations brought about by depressions, wars, and various other conjunctural events are not generally symmetric. Smoothing techniques tend to split the data into symmetric "ups and downs", which means they generally misrepresent the actual deviations from the trend. For instance, in the case of the Great Depression with its sharp collapse and protracted trough in output, the distortion would be quite significant. The oil price shocks of the 1970s would present similar difficulties.

An alternate approach is to focus on a measure of capacity, because we know that cycles, as well as conjunctural events such policy changes, depressions and wars, will be reflected in capacity utilization³. The problem of estimating economic capacity would be relatively simple if one could accept the widely held (neoclassical) assumption that, except for downturns associated with the short (3-5 yr) cycle, capitalist economies generally operate at normal capacity. Indeed, this is the premise of the well-known Wharton method, which defines capacity as the peak output achieved in each business cycle or conjunctural fluctuation. The implicit assumption is that all short-run peaks in output represent the same level (100%) of capacity utilization (Hertzberg, et

Hence any shock to output will be reflected in a
~~investment only effects a net change in capacity; so that~~ *total*
³ ~~The fast process will produce fast fluctuations in output, and since capacity is relatively slowly changing in the short run, it will also produce corresponding fast fluctuations in capacity utilization. On the other hand, over the medium process both the average output and average capacity levels generated in the fast process will themselves fluctuate around each other. Hence, here we will get a corresponding medium cycle in capacity utilization.~~

al, 1974; Schnader, 1984), which automatically excludes the possibility of medium and long term variations in capacity utilization.

A second group of measures tries to get around this problem by relying on economic surveys of operating rates, as in those by the Bureau of Economic Analysis (BEA) and the Bureau of the Census. Here, firms are typically asked to indicate their current operating rate, i.e. their current rate of utilization of capacity. The difficulty with such surveys is that they do not specify any explicit definition of what is meant by capacity. Thus the respondents are free to choose between various measures of capacity, and the analysts who use this data are free to interpret them in manners consistent with their own theoretical premises. A typical case in point is the widely used Federal Reserve Board (FRB) measure of capacity utilization in manufacturing. It begins with a preliminary estimate of capacity by using two different surveys, one by McGraw-Hill (recently discontinued), and one by the Bureau of the Census. The Federal Reserve first combines them in some manner whose details it does not make public. Yet it frequently concludes that the resulting estimates of capacity utilization are not plausible even from own point of view, so it further operates on the combined capacity measures to smooth them out, using regressions on the capital stock and on time (Shapiro 1989, pp. 185-188). Various other adjustments are also made so as to "move the capacity estimate from a peak engineering concept toward an economic concept" consistent with its underlying theory. It is one of the stated goals of these adjustments that the resulting measure of capacity utilization rate is not "chronically below 'normal' capacity utilization" (Shapiro 1989, pp.187-188). *In other words, just as in the case of the Wharton method, the operative premise here is that the economic system generally operates at, or near, full capacity.*

6

A third procedure, such as that employed by the IMF, estimate potential output by means of fitted production functions. As has been often pointed out, a production function represents the *optimal* output which can be produced by given *fully utilized* capital and labor inputs (Fisher 1969). Since actual capital and labor cannot be assumed to fully utilized at any moment (this being the problem under investigation), this method requires some adjustment of the inputs. Thus potential output is estimated using a labor input defined by the natural rate of unemployment and a capital input defined by the trend level of total factor productivity for that particular labor input (De Masi, 1997). Needless to say, this requires theoretical faith not only in the much criticized notion of an aggregate production function (McCombie 2000-2001; Felipe and Fisher 2003; Shaikh 2004) but also in the existence of a natural rate of unemployment. We will return to this issue in the next section, when we compare our own measures of capacity utilization to those of the IMF.

A fourth type measure sidesteps the difficulties inherent in the first two by attempting to directly measure the rate of capacity utilization. In a now classic study, Foss (1963) showed that it is possible to estimate capacity utilization by measuring the utilization rate of the electric motors used to drive capital equipment. Foss's initial estimates for selected years were subsequently developed into an annual series by Jorgenson and Grilliches (1967) and then improved and extended by Christensen and Jorgenson (1969) to cover the period from 1929-1967, and by Shaikh (1992) to cover the period from 1909-1928. But there exists a major obstacle to the forward extension of this series: namely, that the data on the installed capacity of electric motors, which is crucial to the construction of the series, was dropped after the 1963 Census. Shaikh showed that direct survey data available from McGraw-Hill yielded a measure of capacity utilization that was very similar to that derived from data on electric motor use, in their

periods of overlap. This allowed him to splice the two series together to create a complete capacity utilization series from 1947-1985 which differed significantly from the standard Federal Reserve Board (FRB) measure (Shaikh 1987), particularly in terms of longer run patterns such as the Vietnam War buildup during the 1960's, and post-Reagan profit boom from 1982 onward. Unlike the FRB measure, Shaikh's measure is neither symmetric nor stationary over the long run, and it exhibits much greater fluctuations. Conversely, the capacity-output ratio it yields has a much smoother trend than that derived from the FRB measure. Further detail is provided in Shaikh (1987, 1992, 1999) and in section III of this paper, and additional discussion can be found in Winston (1974), Gabish and Lorenz (1989, pp. 26-40), and Tsaliki and Tsoulfidis (1999).

II. The Theoretical Framework

Our framework consists of an identity and two behavioral equations. The identity can be stated as $Y(t) = (Y/Y^*) \cdot (Y^*/K) \cdot K$, where Y = output, Y^* = economic capacity, K = capital stock. If we now define $\check{v}$ = capital-capacity ratio = (Y^*/K) , and $\check{u}$ = capacity utilization rate = (Y/Y^*) , then

Switch to lower case for log

$$1) \log Y(t) = \log K(t) - \log v(t) + \log u(t)$$

Since output (Y) and capital stock (K) are observed variables, to make the preceding identity into a model we need to specify capacity utilization (u) and the capital-capacity ratio (v). On the side of the former, we assume that output fluctuates around capacity over the long run, so that the actual rate of capacity utilization (u_t) fluctuates around some desired or normal rate of capacity

utilization ($u^* = 1$)⁴. This says that firms are able to maintain some correspondence between economic capacity and actual output over the long run, which is consistent with classical, Marxian, Harrodian and neoclassical theory. Nothing in particular is implied about the corresponding utilization of labor supply (Winston, 1974; Kurz, 1986). In log terms, with $e_u(t)$ representing a random error term, this implies

$$2) \log u(t) = e_u(t)$$

Our second behavioral assumption consists of a general specification of technical change in which the capital-capacity ratio changes over time, partly in response to autonomous technical change (coefficient b_1) and partly in response to embodied technical change which itself depends on the rate of capital accumulation (coefficient b_2). Letting g_v = growth rate of the capital-capacity ratio and g_K = growth rate of the capital stock, we have $g_v = b_1 + b_2 \cdot g_K$. In log terms, with an added random error term $e_v(t)$, this yields equation 3 below. *present this*

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$$3) \log v(t) = b_0 + b_1 \cdot t + b_2 \cdot \log K(t) + e_v(t)$$

Equations 1-3 form a general model of the relation of output to the capital stock in the face of technical change and fluctuating capacity utilization. Combining these three equations then yields

$$4) \log Y(t) = a_0 + a_1 \cdot t + a_2 \cdot \log K(t) + e(t)$$

⁴ Although we do not attempt it here, it would be possible to allow for a changing normal rate of capacity utilization

4/3/95

where $a_0 = -b_0$, $a_1 = -b_1$, $a_2 = 1 - b_2$, and the error term $e(t) = e_u(t) - e_v(t)$.

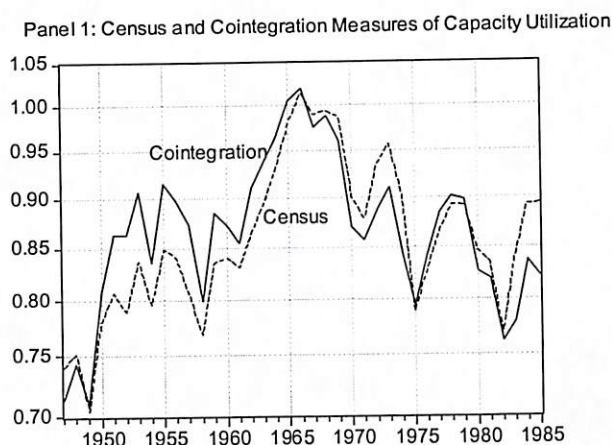
Equation 4 implies that ^{the linear trend is stationary} $\log Y$ and $\log K$ ~~are~~ ^{are} cointegrated, ~~up to a possibly a linear~~ ^{has LR value could} deterministic trend in the actual data. Moreover, from equation 2, the long run value of actual output (Y) is capacity output (Y*). With an estimate of capacity in hand, we can then derive the rate of capacity utilization $u = Y/Y^*$ and the capital-capacity ratio $v = K/Y^*$. ^{The question is, how} does one ~~investigate~~ ^{investigate} the long run relation between $\log Y$ and $\log K$? ^{← coint. ARDL}

III. An Application to the US Manufacturing Sector

As a benchmark test of our methodology, we utilized the above cointegration framework to derive an econometric measure of the capacity utilization rate for the US manufacturing sector using data on natural logs of real manufacturing GDP and real net capital stock (see the Data Appendix for details). This was then compared to Shaikh's census-based measure, previously discussed at the end of section I. As Panel 1 shows, the close correspondence between the patterns of the two very different techniques provides a sense of the power of our methodology⁵. The close fit between these two independent methods is quite striking.

in the light of changing real wages and relative material prices. This is because the point at which a plant is most profitably utilized can vary with changes in input prices (Kurz 1986).

⁵ Cointegration in levels of variables requires some choice of scale, which is usually accomplished by arbitrarily setting the cointegration coefficient of the first variable equal to one. An independent measure of the scale of economic capacity would be ideal. In the present case in which we are comparing our econometric measure of capacity utilization to an independent census-based measure, we can use the average level of the latter to define the scale of the former. But when making comparisons across countries for which no such independent measures are available, and in which some cointegration regressions were in log-levels and others in growth rates (first



IV. Empirical measures of capacity and technical change in OECD countries

X Extended
Perron world
Tables

Data on business sector real output and capital stock was obtained from the *OECD Economic Outlook 71* (June 2002) database. Given the relatively small sample sizes and the fact that the unit root tests have low power in such cases, we follow the literature by utilizing three different tests to gauge the order of integration⁶. In general, our final determination of the order of integration of the logs of each variable was based on agreement among at least two of the tests.

The levels of the natural logs of both output and capital stock were $I(1)$ for Australia, Canada, France, Norway, Sweden, UK, and US, whereas they are $I(2)$ for Austria, Germany, Japan, and Spain. In the former group of countries the Johansen cointegration test was carried out on the level of the variables, whereas in the latter group it was carried out on their first differences. In light of our previous discussion concerning scale (see footnote 5), we scaled the level of capacity by equating it to actual output in the year

differences of log-levels), it is somewhat easier to define scale by using the highest growth rate peak over the period of observation. This is the procedure we adopt in the next section.

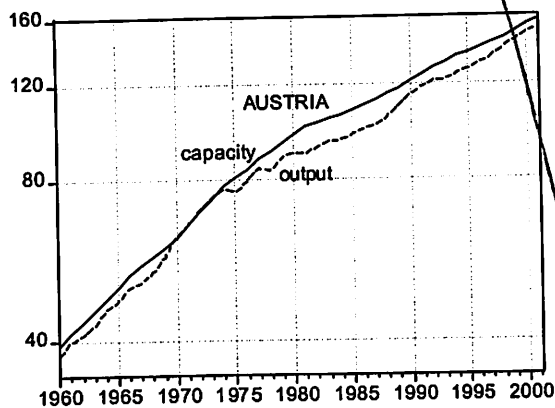
⁶ We used the augmented Dickey-Fuller (ADF), the Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and the Ng-Perron (NP) tests.

with the peak growth rate. The cointegration tests parallel the form of equation 4, in that they assume a linear deterministic trend in the actual data and an intercept in the cointegrating equation. Further details are provided in the data appendix.

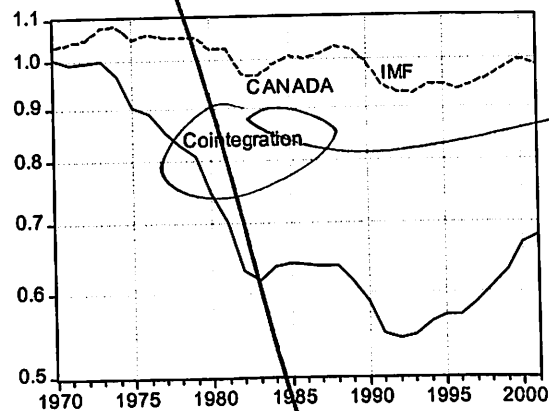
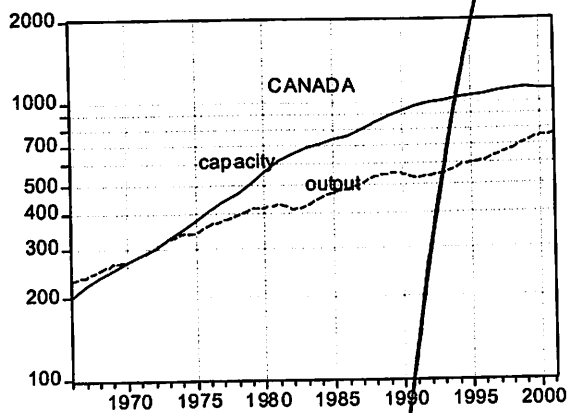
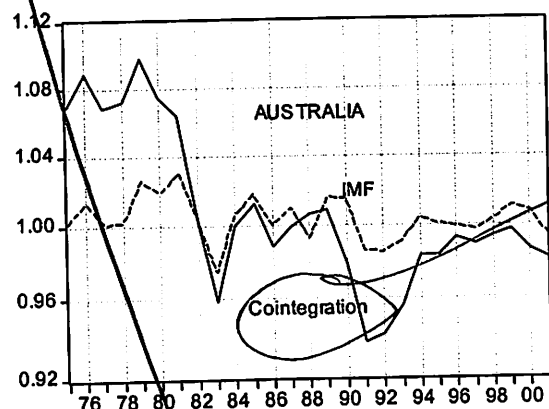
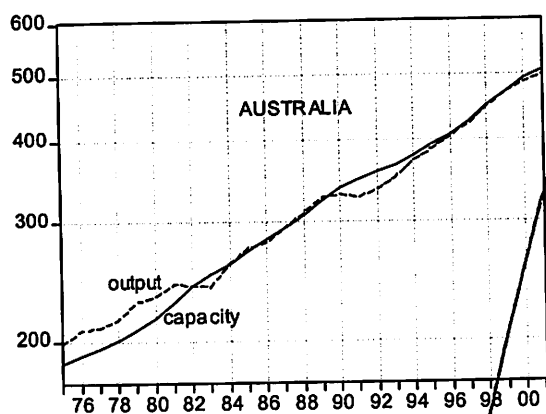
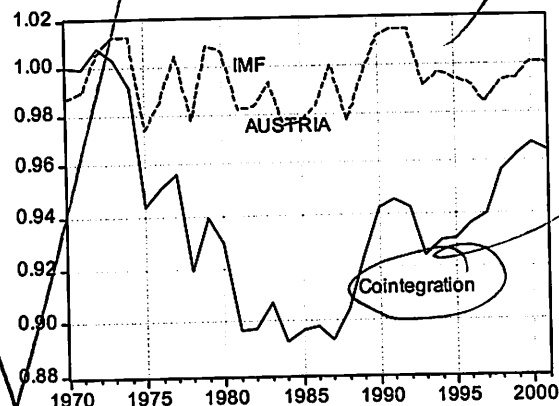
The succeeding set of graphs in Panel 2 display our estimates of capacity and the rate of capacity utilization, the latter in comparison to corresponding estimates from the IMF. Three features are notable. First, our measures of capacity do not simply “thread-through” the actual level of output, as smoothed and filtered methods do. This is generally so, but is most evident in the case of countries such as Austria, Canada, and the UK. Secondly, the measures of capacity utilization derived from our cointegration method are generally very different from those provided by the IMF. It will be recalled from Section I that the IMF defines potential output via an aggregate production function in which labor is assumed to be employed at the natural rate and capital is assumed to be utilized at the rate corresponding to the trend level of factor productivity. On the other hand, we measure potential output as the component of actual output that co-varies with the capital stock over the long run. In principle, both measures should filter out short run movements and highlight long run ones. Yet our measure of capacity utilization tends to exhibit a wider range of variations than those from the IMF. This is because our cointegration measure of capacity, which we derive explicitly, is smoother than that implicit in the IMF calculations.

Panel 2: Output, Capacity, and Capacity Utilization Rate

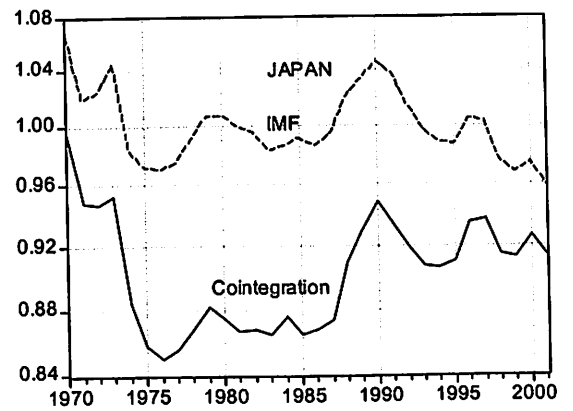
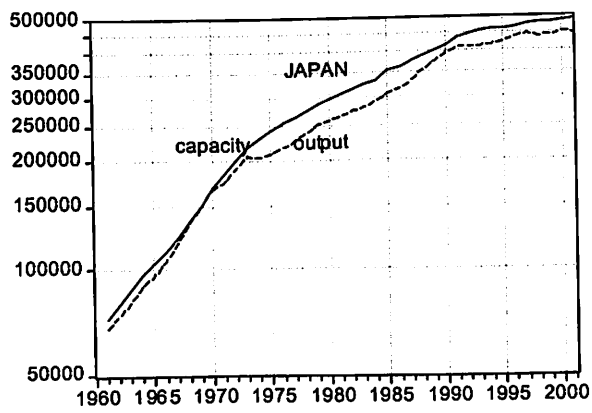
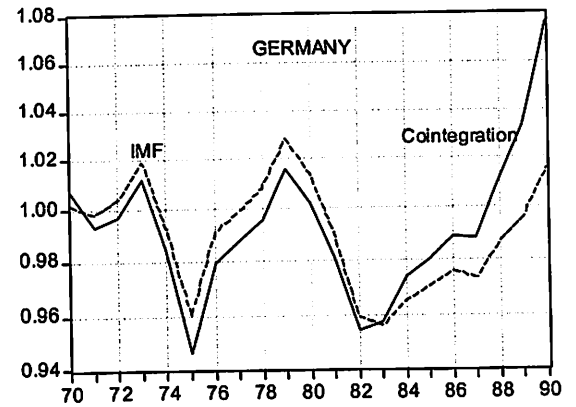
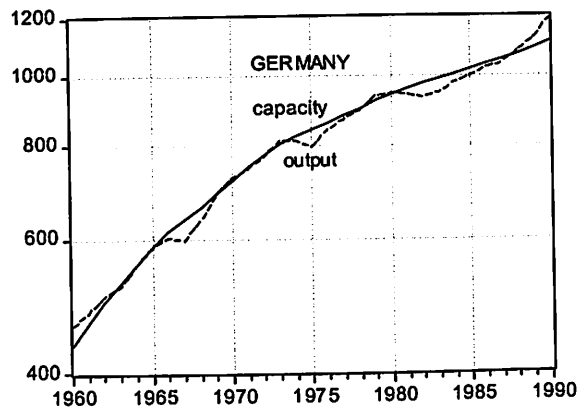
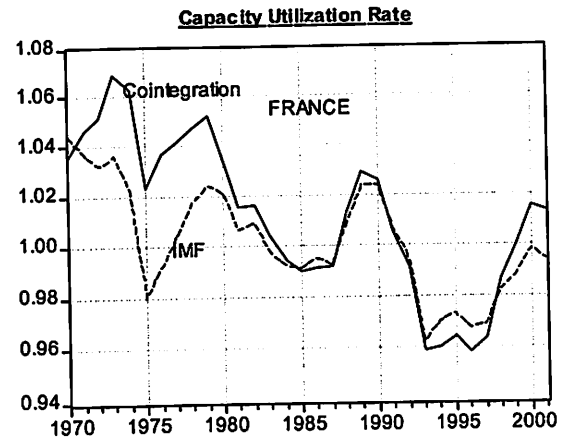
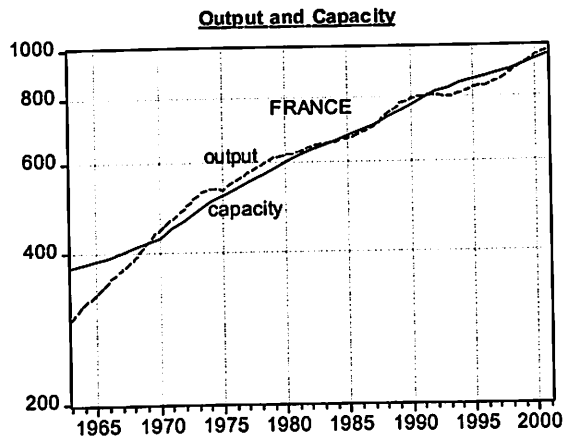
Output and Capacity



Capacity Utilization Rate

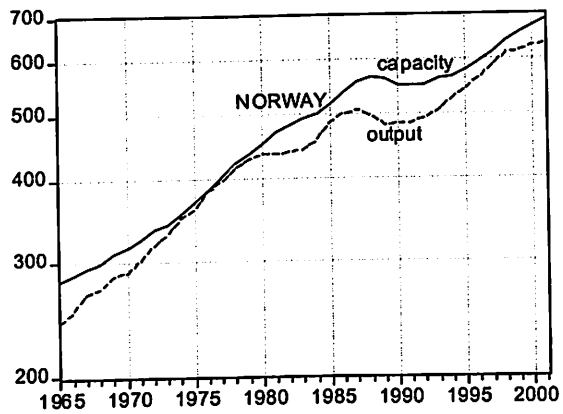


Panel 2(continued):

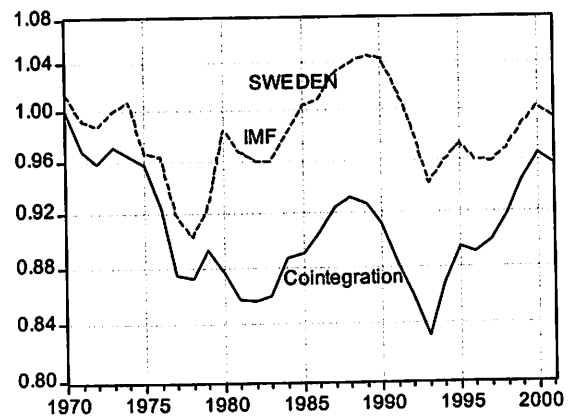
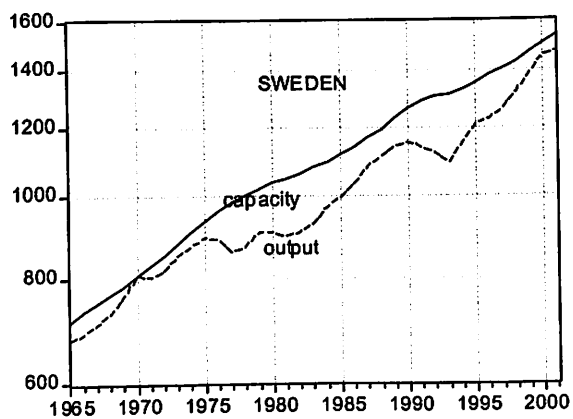
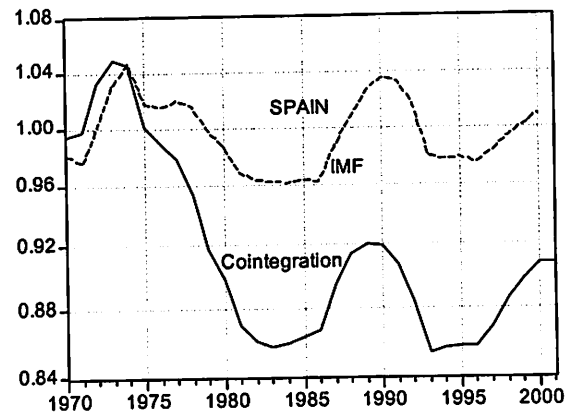
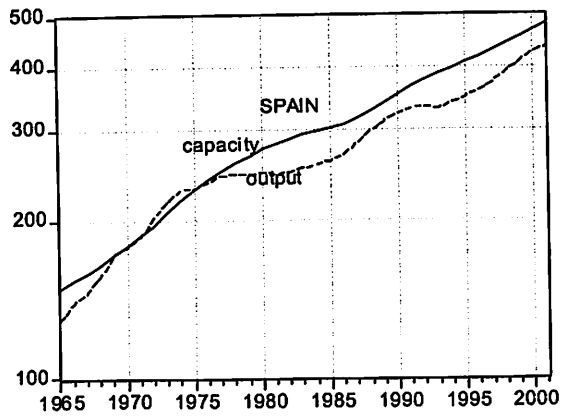
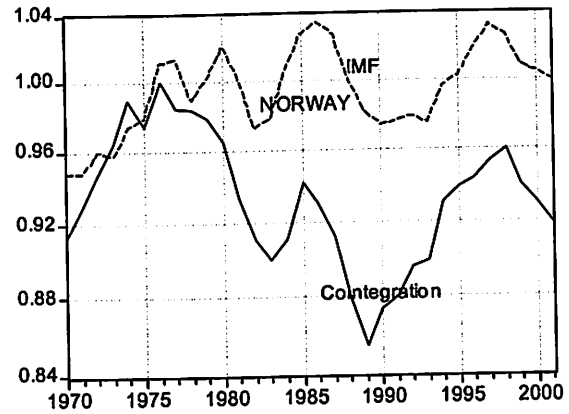


Panel 2 (continued):

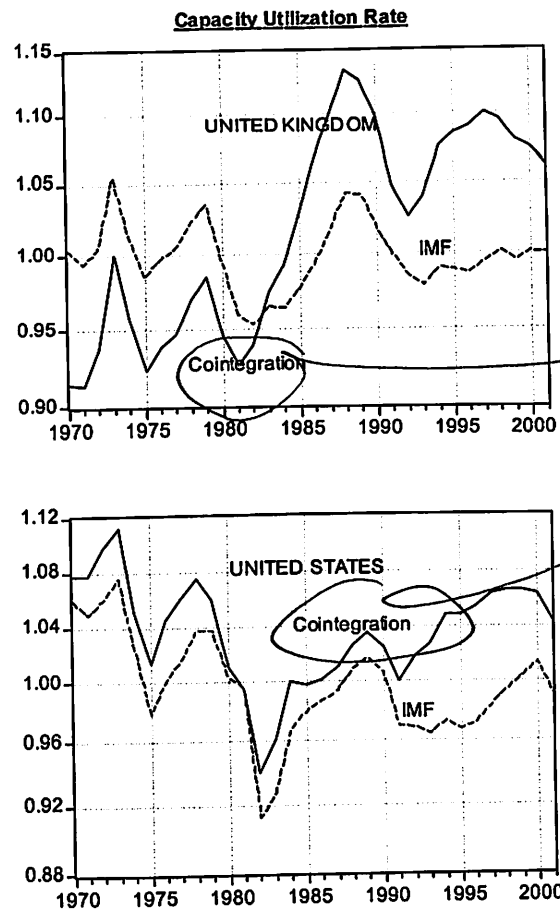
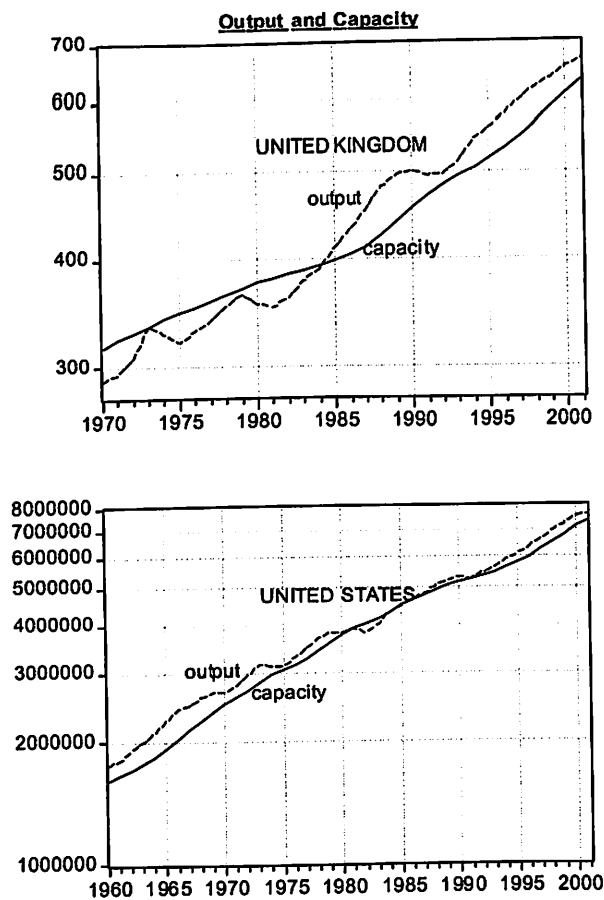
Output and Capacity



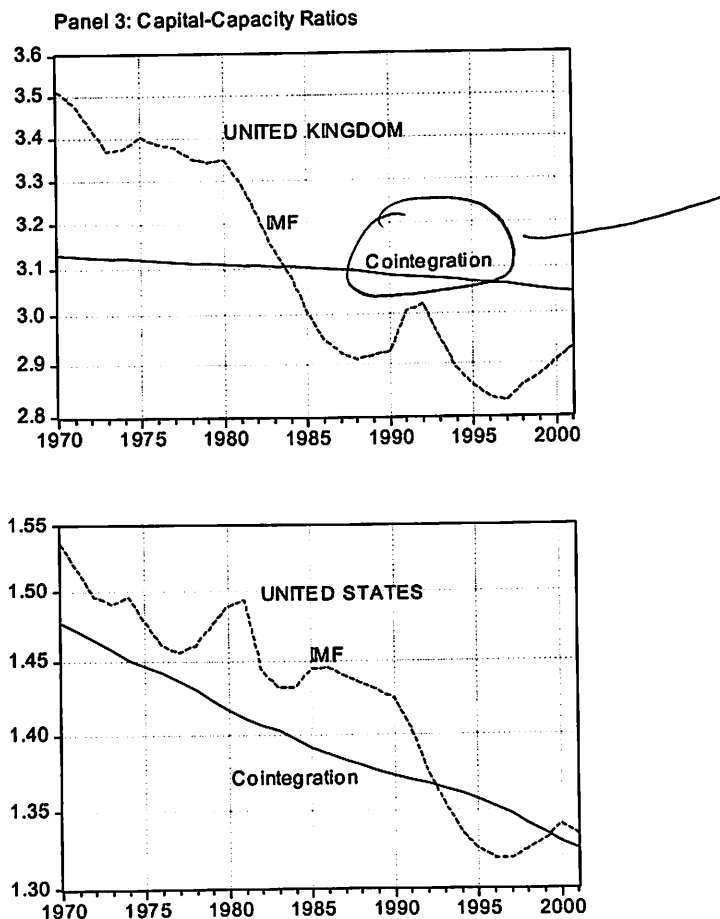
Capacity Utilization Rate



Panel 2 (continued):



Finally, since technical change is generally embodied at the margin through new investment, one would expect the capital-capacity ratio (the reciprocal of the “productivity of capital”) to respond fairly smoothly to technical change. Our measure always exhibits this property, but it turns out that the IMF one often does not. Panel 3 displays the two such cases, for the U.S. and the U.K. Considerations of space prevent us from exhibiting all of the relevant graphs.



IV. Conclusion

In this paper, we provide a simple general method for estimating economic capacity. The intuitive idea is that economic capacity (potential output) is the aspect of output that co-varies with the capital stock over the long run. We show that this notion can be derived from a general model that allows for a changing capital-capacity ratio in response to partially exogenous, partially embodied, technical change. Our method has several advantages. It only requires data on output and capital stock, which is widely available across countries, and even across

industries. It also closely replicates a previously developed census-based measure of U.S. manufacturing capacity-utilization. Of particular interest is that our measures of capacity are quite different from standard ones based on aggregate production functions, such as those provided by the IMF. The two yield quite different patterns for capacity utilization and for the capital-capacity ratio (the reciprocal of the "productivity of capital"). In the latter instance, our measure always yields a smooth capital-capacity ratio, while the IMF measure often does not.

Several extensions of this paper are possible. The cointegration method itself could be easily applied at industry levels, since it only requires data on output and capital stock. In addition, the method itself could be deepened by testing for structural breaks in the cointegration equation. Finally, the method could be applied to variables other than output, such as employment and profit, so as to assess the long term trends of technical change and profitability, respectively.

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References

- BEA web site (Bureau of Economic Analysis, U.S. Department of Commerce: www.bea.gov).
- Christensen, L. R. and Jorgenson, D. W. (1969) The Measurement of US Real Capital Input, 1929-1967, *Review of Income and Wealth*, vol. 15, no. 4.
- _____. (1976) Measuring Economic Performance in the Private Sector, *Studies in Income and Wealth*, vol. 28. New York: National Bureau of Economic Research.
- De. Masi, P. (1997) IMF Estimates of Potential Output: Theory and Practice, *IMF Working Paper*, no. 97/177.
- Enders, W. (1995) *Applied Econometric Time Series*. John Wiley & Sons, Inc.
- Fisher, F.M. (1969) The Existence of Aggregate Production Functions, *Econometrica*, Vol. 37, No. 4, October, pp. 553-77.
- Felipe, J. and F.M. Fisher. (2003) Aggregation in Production Functions: What Applied Economists Should Know, *Metroeconomica*, Vol. 54, May, pp. 208-262.
- Foss, M. F. (1963) The Utilization of Capital Equipment: Postwar Compared with Prewar, *Survey of Current Business*, vol. 43. pp. 8-16.
- Gabisch, G. and Lorenz, H - W. (1987) *Business Cycle Theory: A Survey of Methods and Concepts*. Berlin: Springer-Verlag.
- Garegnani, P. (1979) Notes on consumption, investment, and effective demand: a reply to Joan Robinson, *Cambridge Journal of Economics*, 3(3), 181-187

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- Gillman, J. (1958) *The Falling Rate of Profit, Marx's Law and Its Significance to Twentieth Century Capitalism*. New York: Cameron Associates.
- Hertzberg, M. R., Jacobs, A. I. and Trevathan, J. E. (1974) The Utilization of Manufacturing Capacity, 1965-73, *Survey of Current Business*.
- Jorgenson, D.W. and Z. Grilliches. (1967) The Explanation of Productivity Change, *Review of Economic Studies*, Vol. 34.
- Kurz, H. (1986) Normal Positions and Capital Utilization, *Political Economy: Studies in the Surplus Approach*, vol. 2, number 1, pp. 37-54.
- Marris, R. (1964) *The Economics of Capital Utilization: A Report on Multiple-Shift Work*. (Cambridge: Cambridge University Press).
- McCombie (2000-2001) The Solow residual, technical change, and aggregate production functions, *Journal of Post Keynesian Economics*, Winter 2000-01, 267-97. *OECD Economic Outlook*, 71, June.
- Perlo, V. (1968) Capital-Output Ratios in Manufacturing, *Quarterly Review of Economics and Business*, pp. 29-42
- Raddock, R. D. and Forest, G. R. (1978) *Federal Reserve Measures of Capacity and Capacity Utilization*. (Board of Governors of the Federal Reserve System).
- Ragan, J. M. (1976) Measuring Capacity Utilization in Manufacturing, *Federal Reserve Board New York Quarterly Review*.
- Rost, R. F. (1980) New Federal Reserve Measures of Capacity and Capacity Utilization, *Federal Reserve Bulletin*.
- Schnader, M.H. (1984) Capacity Utilization, *The Handbook of Economic and Financial Measures*, Frank J. Fabozzi and Harry I. Greenfield (editors), Illinois: Dow-Jones Irwin.
- Shaikh, A. (1987) *The Current Economic Crisis: Causes and Implications, The Imperiled Economy*, Book 1, URPE, N.Y.
- _____ (1992) The Falling Rate of Profit and Long Waves in Accumulation: Theory and Evidence, in Alfred Kleinknecht, Ernest Mandel, Immanuel Wallerstein (eds.), *New Findings in Long Wave Research*, London: Macmillan.
- _____ (1999) Notes on Capacity Utilization Measures. *Mimeo*, New School University.
- _____ (2004 forthcoming) Nonlinear Dynamics and Pseudo-Production Functions, *Eastern Economics Journal*, Special Issue on Production Functions. Spring.
- Shapiro, M. (1989) Assessing the Federal Reserve's Measures of Capacity and Utilization, *Brookings Papers on Economic Activity*, 1, pp. 181-241.
- Tsaliki, P. and L. Tsoulfidis (1999) Capacity Utilization in Greek Manufacturing, *Modern Greek Studies Yearbook*, Vol 9, pp. 127-42.
- van Duijn, J.J. (1983) *The Long Wave in Economic Life*. (London: George Allen and Unwin).
- Weisskopf, T. (1979) Marxian Crisis Theory and the Rate of Profit in the Postwar U.S., *Cambridge Journal of Economics*, no. 4.
- Winston, G. C. (1974) The Theory of Capital Utilization and Idleness. *Journal of Economic Literature*, 12, 4, pp. 1301 - 1320.
- Wolff, Edward N. (1986) The Productivity Slowdown and the Fall in the U.S. Rate of Profit, 1947-76. *Review of Radical Political Economy* Vol. 18, No. 1&2, 87-109.

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wrong
title
this is
against
the demand!

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144

DATA APPENDIX

As is well known, the power of unit root tests is rather low, especially when applied to relatively small sample sizes such as the current one. We have therefore decided to carry out three tests: the Augmented Dickey Fuller (ADF), the Kwiatkowski, Phillips, Schmidt, and Shin (KPSS), and the Ng – Perron (NP) tests. For both the ADF and the NP tests the null hypothesis is the presence of a unit root while that for the KPSS test is that the variable is stationary. We decided on the order of integration of a variable on the basis of at least two of the tests. A decision based on at least two tests made us reasonably confident of our final specification with regard to the number of unit roots in each variable.

In the case of the manufacturing sector, we used natural logs of real manufacturing GDP and real net capital stock (BEA web site). Real output was $I(1)$ at 1% levels in all three tests, whereas real capital stock was $I(1)$ at levels ranging from 1%, 5%, and 10% in the three respective tests. Reporting our coefficients in a standard regression form $\ln Y = f(\ln K)$, the constant is 4.786 and the coefficient of $\ln K$ is 0.652, with a t-statistic of 22.52.

For the OECD country data, our unit root test results are summarized in Table 1.

TABLE 1: Unit Root Tests on Natural Logs of Real Output and Capital Stock

Country	Order of Integration, Type of Test, and Significance Level	
	Real Output	Real Capital Stock
Austria	I(2) KPSS*** & NP*	I(2) KPSS* & NP*
Australia	I(1) KPSS*** & NP*	I(1) KPSS** & NP*
Canada	I(1) ADF*, KPSS**, & NP*	I(1) ADF*, KPSS**, & NP*
France	I(1) ADF*, KPSS**, & NP*	I(1) ADF*, KPSS*, & NP*
Germany	I(2) ADF*, KPSS***, & NP*	I(2) ADF*, KPSS**, & NP*
Japan	I(2) ADF*, KPSS*, & NP*	I(2) ADF*, KPSS*, & NP*
Norway	I(1) KPSS** & NP*	I(1) KPSS* & NP*
Spain	I(2) ADF* & NP*	I(2) ADF* & NP*
Sweden	I(1) KPSS** & NP*	I(1) KPSS* & NP*
UK	I(1) KPSS*** & NP**	I(1) KPSS** & NP*
US	I(1) ADF*, KPSS***, & NP*	I(1) ADF*, KPSS**, & NP*

* = significance at 1%, ** = significance at 5%, and

*** = significance at 10%

The Johansen cointegrating test was carried out for all the countries. In the case of those countries for which the variables are I(1), the regression was run on the levels of the variables. For those countries for which the variables are I(2), their first differences were I(1), so the cointegration test was performed on these. With regard to the second group of countries, the

cointegration relationship allowed us to derive the growth rate of the capacity utilization rate. It was then assumed that in some peak year of output growth, the capacity utilization rate was at the normal level and was set equal to one. This base year value and the growth rate of the rate of capacity utilization were used to derive the level of this variable.

The regressions were run by assuming a linear deterministic trend in the actual data and an intercept but no trend in the cointegrating equation. Table 2 below shows the coefficients on the right-hand-side of the long – run cointegration equations (t – statistics are in parenthesis). We report our results in standard regression form, i.e. with $\ln Y$ or its first difference on the left hand side, and the constant and $\ln K$ or its first difference on the right hand side.

TABLE 2: Cointegration Results (Natural logs of real output, capital stock)

	Constant	Capital Stock Coefficient
Country		
*Austria	-0.014	0.889 (7.27)
Australia	-0.145	0.955 (26.66)
Canada	-4.25	1.545 (6.7)
France	1.04	0.766 (31.56)
*Germany	-.0035	0.672 (7.04)
*Japan	-0.017	0.888 (12.52)
Norway	-0.71	1.014 (43.08)
*Spain	0.002	0.658 (3.14)
Sweden	0.434	0.818 (39.62)
UK	-1.439	1.043 (15.87)
US	-2.088	1.112 (38.63)

* For these countries the estimated long-run relationship is between output and the capital stock growth rates.

Appendix I

The following procedure, entailing the use of OLS, was followed to derive the long-run relation between output and the capital stock:

1. Following the methodology derived by Perron (1989) we carried out unit root tests using structural breaks (Enders, 1994, pp. 245-248) in order to make sure that none of the variables were $I(2)$. Pesaran's bounds test cannot be used with such variables. For this purpose, Perron's unit root tests were applied to the first differences of the two variables. We found in all cases that the first differences of the variables were $I(0)$ implying that the levels are either $I(1)$ or $I(0)$. This allowed us to carry out the bounds test since, in contrast to the Johansen method, this method is applicable irrespective of whether the underlying regressors are purely $I(0)$, purely $I(1)$ or mutually co-integrated.
2. Since the point of departure of the method derived by Pesaran et al (2001) is the conditional ECM, we first used this equation to consider the possibility of including dummy variables. A recursive least squares estimate of this equation was made and the recursive residual was subjected to a battery of tests to investigate the possibility of parameter instability. We also checked the squared values of the residuals to identify any significantly large jumps which usually imply an 'outlier' (Patterson, 2000, p. 195). Appropriate dummies were selected on the basis of these two methods.
3. Time trends were included in the conditional ECM if their t statistics were significant at the 5% level.

Not really
we ended up
with
outliers?

Australia
Austria
Belgium
Canada
Denmark
France
Great Britain
Japan
Luxembourg
Netherlands
Portugal
Sweden
Spain
United States

DUMMIES

The following is the list of dummies from the best ADL equations in CAPACITY ESTIMATES 3.WF1:

aus: d80to82
aut: d73 d81 d83
bel: d75 d80 d93
can: d74 d91
den: d75 d81
esp: d82 d95
fra: d74 d80 d89
gbr: d73 d81 d84
jpn: d74
lux: d74 d8182
nld: d75 d76 d81 d82
prt: d68 d73
swe: d83 d92
us: d74 d80to82

STABILITY TESTS FOR "GOOD RESULTS" COUNTRIES

Below is listed for each country periods of instability and the test used. The recursive least squares test was carried on the best ADL equations (see below). Note that Eviews only plots the output for the years after the highest-year dummy.

Australia: CUSUM SQ (1988-89), CUSUM (persistent instability 1988-2000), 1-step (1988), N-step (1988), recursive resids (1988)
Belgium: CUSUM SQ (1998-99)
Canada: recursive resids (1997), 1-step (1997), N-step (1995, 1996, 1997), recursive coeff (1996)
Denmark: recursive resids (1989, 1995), CUSUM SQ (1988), 1-step (1989, 1995), N-step (1986, 1987, 1988, 1989), recursive coeff (1989)
France: 1-step (1995)
Great Britain: CUSUM SQ (1989-1991), 1-step (1986, 1989), recursive coeff (1992?)
Japan: recursive resids (1988, 1990), 1-step (1988, 1990), N-step (1985, 1986, 1987), recursive coeff (1987)
Portugal: 1-step (1996)
Spain: recursive resids (1998), 1-step (1998), N-step (1996, 1997, 1999)

BAD RESULTS

Ireland, Italy, Norway, Greece: Capacity persistently above output
Switzerland: Serial correlation for all lag combinations in conditional ECM
Finland: no long-run relationship found

BEST MODELS

Japan: eqadljpnstarb
US: eqadlustara
Australia: eqadlausstarb (Can this fit be improved? Average UAUSB is 1.009)
Austria: eqadlautstarc
Belgium: eqadlbelstar
Canada: eqadlcanstarb

Our Capacity Utilization Model

6/8/08

1. Our technical change function ($v \equiv \log V$, $k \equiv \log K$)

1) $\Delta v = a_1 + a_2 \Delta k + \varepsilon_v$

1)' $\boxed{v = a_1 + v_{-1}} + a_2 \Delta k + \varepsilon_v$
 $\rightarrow$ unit root component

- Banerjee, et al, (1993) points out that if $\Delta k (\equiv \Delta \log K)$ ^{is both exogenous and I(0)}, then v is a unit root process of the form

2) $\boxed{v = a_1 + v_{-1} + \Delta \phi}$ where $\phi \equiv a_1 \Delta k + \varepsilon_v$ [Banerjee, et al, 1993: pp 84-86, 87]

- Of course, as a unit root process, it can then also be represented as

3) $v = a_1 t + \sum_{i=1}^t \phi_i$ [Banerjee, 1993, p. 87]

But $\sum \phi_i = a_1 \Delta k_1 + a_2 \Delta k_2 + \dots + \sum \varepsilon_{vi}$
 $= a_2 [k_1 - k_0 + k_2 - k_1 + k_3 - k_2 + \dots + (k_t - k_{t-1})] + \sum \varepsilon_{vi}$
 $= a_2 k_t - (a_1 k_0) + \sum \varepsilon_{vi}$

✓ $\therefore$ 4) $\boxed{v = a_0 + a_1 t + a_2 k + \sum \varepsilon_{vi}}$ = where $a_0 = -a_1 k_0$

4)' $\boxed{v = a_1 (t - k_0) + a_2 k + \sum \varepsilon_{vi}}$ $\rightarrow$ RW
 $\rightarrow$ Nonlinear LS

2. Significance of these two representations

If $y = k + u - v$ combined with eq 2 or eq 4, gives

5) $y \equiv k - v + u$

2) $v = a_1 + (v_{-1}) + a_2 \Delta k + \varepsilon_v$

6) u is trendless

5) $y \equiv k - v + u$

4) $v = a_0 + a_1 t + a_2 k + \sum \varepsilon_v$

6) u is trendless

← This is fine for a Kalman filter, but not for a regression, since v_{-1} on RHS is endogenous

Here $(\varepsilon_v) = iid$, so $\sum \varepsilon_v$ is MA(1)

Logic of ARDL Method

ECM to do P test
ARDL to get LR coeffs

5/2/25

1. Start with initial test ECM

$k=1$, since we have only one independent variable (dummies do not count, Pesaran 2001, 1115-19)

on first differences.

4 lags $F < F_{\alpha}$ X

3 " " " X

2 lags $F > F_{\alpha}$ ✓

2. Baseline = 2 lags in differences $\rightarrow$ 3 lags in levels

$\rightarrow$ RUN IT AS AN ADL (i.e. in levels), not an ECM

3. MIN AIC

EViews

(i) Prune least significant coefficients $\rightarrow$ MIN AIC

(ii) Re-run MIN AIC in levels

Microfit

Runs all possible combinations of lag lengths in levels $\rightarrow$ MIN AIC

MIN AIC in levels

LR coefficients

ecm =

Re-run in ECM-form using ecm as a variable, [and zero ($\equiv$ dc) as a variable] in EViews & / or Microfit.

5/2/05

1

1. Start with ECM with up to four lags, but then the F is too low: $F^* = 4.94, 5.73$

EQARDL00

Dependent Variable: D(LNYRMFG)

Method: Least Squares

Date: 04/26/05 Time: 16:25

Sample (adjusted): 1952 2000

Included observations: 49 after adjustments

Note that if $F < (F_1^*, F_2^*)$, then
we cannot reject the null that there is
no LR relationship (see Microfit manual
pp. 305-306)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.737681	0.678118	2.562505	0.0146
LNYRMFG(-1)	-0.292392	0.148830	-1.964599	0.0570
LNKRMFG(-1)	0.167513	0.102121	1.640335	0.1094
D(LNKRMFG)	3.017991	0.683745	4.413913	0.0001
D(LNKRMFG(-1))	-3.570826	0.915512	-3.900357	0.0004
D(LNKRMFG(-2))	1.719296	1.107184	1.552856	0.1290
D(LNKRMFG(-3))	0.778861	1.081575	0.720117	0.4760
D(LNKRMFG(-4))	-1.614447	0.689139	-2.342702	0.0246
D(LNYRMFG(-1))	-0.100032	0.175956	-0.568507	0.5731
D(LNYRMFG(-2))	0.221876	0.168091	1.319970	0.1950
D(LNYRMFG(-3))	-0.293773	0.164189	-1.789237	0.0818
D(LNYRMFG(-4))	-0.117420	0.142165	-0.825944	0.4141
R-squared	0.504889	Mean dependent var	0.020146	
Adjusted R-squared	0.357694	S.D. dependent var	0.045774	
S.E. of regression	0.036685	Akaike info criterion	-3.563983	
Sum squared resid	0.049795	Schwarz criterion	-3.100680	
Log likelihood	99.31757	F-statistic	3.430070	
Durbin-Watson stat	2.077185	Prob(F-statistic)	0.002347	

$< F^* = 4.94, 5.73$

2. Lower Lag length until we get an $F > F^*$, so we can reject the null of no LR relationship

EQARDL01

Dependent Variable: D(LNYRMFG)

Method: Least Squares

Date: 04/26/05 Time: 16:28

Sample (adjusted): 1950 2000

Included observations: 51 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.461406	0.524184	2.787966	0.0079
LNYRMFG(-1)	-0.275185	0.118936	-2.313720	0.0255
LNKRMFG(-1)	0.169137	0.083679	2.021265	0.0495
D(LNKRMFG)	2.809850	0.658856	4.264738	0.0001
D(LNKRMFG(-1))	-3.640514	0.811067	-4.488549	0.0001
D(LNKRMFG(-2))	1.400806	0.624138	2.244386	0.0300
D(LNYRMFG(-1))	-0.095970	0.162618	-0.590156	0.5582
D(LNYRMFG(-2))	0.226738	0.136140	1.665472	0.1031
R-squared	0.498864	Mean dependent var		0.023959
Adjusted R-squared	0.417284	S.D. dependent var		0.048948
S.E. of regression	0.037365	Akaike info criterion		-3.593063
Sum squared resid	0.060034	Schwarz criterion		-3.290031
Log likelihood	99.62310	F-statistic		6.115007
Durbin-Watson stat	1.880434	Prob(F-statistic)		0.000055

$F^* = 4.94, 5.73$

3. Now we prune to minimize AIC

EQECMFTESTEND
 Dependent Variable: D(LNYRMFG)
 Method: Least Squares
 Date: 04/25/05 Time: 17:08
 Sample (adjusted): 1950 2000
 Included observations: 51 after adjustments

X

This is not how Microsoft does it
 They run model in levels, as
 an ADL, then prune by AIC,
 SBC, etc.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.529011	0.507711	3.011579	0.0043
LNYRMFG(-1)	-0.297591	0.111876	-2.660012	0.0109
LNKRMFG(-1)	0.186323	0.077864	2.392929	0.0210
D(LNKRMFG)	2.675018	0.613382	4.361098	0.0001
D(LNYRMFG(-2))	0.251511	0.128544	1.956617	0.0568
D(LNKRMFG(-1))	-3.620587	0.804340	-4.501314	0.0000
D(LNKRMFG(-2))	1.568139	0.551882	2.841436	0.0068
R-squared	0.494805	Mean dependent var	0.023959	
Adjusted R-squared	0.425915	S.D. dependent var	0.048948	
S.E. of regression	0.037087	Akaike info criterion	-3.624211	
Sum squared resid	0.060521	Schwarz criterion	-3.359059	
Log likelihood	99.41739	F-statistic	7.182513	
Durbin-Watson stat	1.977459	Prob(F-statistic)	0.000022	

one way to get

i. This gives us the LR coefficients, from which we have the estimated long run relationship between LNYRMFG and LNKRMFG

$$\boxed{\text{Lnyrmfg}^*} = (1.529011/0.297591) + (0.186323/0.297591) * \text{lnkrmfg} = \boxed{5.138 + .626 \text{ lnkrmfg}}$$

ii. Now generate the residual called "ecm", where

$$\text{ecm} = \text{lnyrmfg} - \text{lnyrmfg}^*$$

[note that these are very close to the "true" LR coefficients estimated from levels relation on next page, which gives $\text{lnyrmfg} = 5.311 + .615 \text{ lnkrmfg}$]

Alternately, we could run the previous (min AIC) ECM in levels instead, and use this to calculate the LR coefficients and the residual "ecm". This is what MICROFIT does.

(see p. 4A from Microfit)

EQECMFTESTEND1

Dependent Variable: LNYRMFG

Method: Least Squares

Date: 04/26/05 Time: 16:44

Sample (adjusted): 1950 2000

Included observations: 51 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.461406	0.524184	2.787966	0.0079
LNYRMFG(-1)	0.628845	0.168055	3.741896	0.0005
LNKRMFG	2.809850	0.658856	4.264738	0.0001
LNKRMFG(-1)	-6.281227	1.280245	-4.906269	0.0000
LNKRMFG(-2)	5.041320	1.306912	3.857429	0.0004
LNKRMFG(-3)	-1.400806	0.624138	-2.244386	0.0300
LNYRMFG(-2)	0.322708	0.176992	1.823295	0.0752
LNYRMFG(-3)	-0.226738	0.136140	-1.665472	0.1031
R-squared	0.986559	Mean dependent var	13.72108	
Adjusted R-squared	0.984371	S.D. dependent var	0.298877	
S.E. of regression	0.037365	Akaike info criterion	-3.593063	
Sum squared resid	0.060034	Schwarz criterion	-3.290031	
Log likelihood	99.62310	F-statistic	450.8689	
Durbin-Watson stat	1.880434	Prob(F-statistic)	0.000000	

This gives us an alternate route to the LR coefficients, from which we have the estimated long run relationship between LNYRMFG and LNKRMFG

$$\ln y_{rmt} = 5.3106 + 0.6146 \ln k_{rmt}$$

ii. Now generate the residual called "ecm", where

$$ecm = \ln y_{rmt} - \ln y_{rmt}^*$$

$$\hat{\phi}_1 = 0.628845 \quad \hat{\phi}_2 = 0.322708 \quad \hat{\phi}_3 = -0.226738$$

$$1 - \sum \hat{\phi}_i = 0.275185$$

$$\hat{\beta}_0 = 2.80985 \quad \hat{\beta}_1 = -6.281227 \quad \hat{\beta}_2 = 5.041320 \quad \hat{\beta}_3 = -1.400806$$

$$\sum \hat{\beta}_i = 0.169137$$

$$\theta = \frac{\sum \hat{\beta}_i}{1 - \sum \hat{\phi}_i} = \frac{0.169137}{0.275185} = 0.61463$$

(Same as coeff of ln k in Microfit, see p. 4B)

$$\hat{c} = \frac{c}{1 - \sum \hat{\phi}_i} = \frac{1.461406}{0.275185} = 5.3106$$

(")

Microfit Manual
pp 393-394

Coeff of ln k

Page 1

Jamee Moudud
2/May/2005 0:0:0

From Microfit

3 lags

ADL form

Autoregressive Distributed Lag Estimates
ARDL(3,3) selected based on Akaike Information Criterion

```
*****
Dependent variable is LNY
51 observations used for estimation from 1950 to 2000
*****
```

Regressor	Coefficient	Standard Error	T-Ratio[Prob]
LNY(-1)	.62884	.16806	3.7419[.001]
LNY(-2)	.32271	.17699	1.8233[.075]
LNY(-3)	-.22674	.13614	-1.6655[.103]
LNK	2.8098	.65886	4.2647[.000]
LNK(-1)	-6.2812	1.2802	-4.9063[.000]
LNK(-2)	5.0413	1.3069	3.8574[.000]
LNK(-3)	-1.4008	.62414	-2.2444[.030]
C	1.4614	.52418	2.7880[.008]

```
*****
R-Squared .98656 R-Bar-Squared .98437
S.E. of Regression .037365 F-stat. F( 7, 43) 450.8687[.000]
Mean of Dependent Variable 13.7211 S.D. of Dependent Variable .29888
Residual Sum of Squares .060034 Equation Log-likelihood 99.6231
Akaike Info. Criterion 91.6231 Schwarz Bayesian Criterion 83.8958
DW-statistic 1.8804
*****
```

Diagnostic Tests

```
*****
* Test Statistics * LM Version * F Version *
*****
```

Test	LM Version	F Version
A:Serial Correlation*CHSQ(1)=	1.6466[.199]*F(1, 42)=	1.4013[.243]*
B:Functional Form *CHSQ(1)=	.014101[.905]*F(1, 42)=	.011616[.915]*
C:Normality *CHSQ(2)=	7.3151[.026]*	Not applicable
D:Heteroscedasticity*CHSQ(1)=	.013297[.908]*F(1, 49)=	.012779[.910]*

```
*****
A:Lagrange multiplier test of residual serial correlation
B:Ramsey's RESET test using the square of the fitted values
C:Based on a test of skewness and kurtosis of residuals
D:Based on the regression of squared residuals on squared fitted values
*****
```

Jamee Moudud
2/May/2005 0:0:0

4B

Error Correction Representation for the Selected ARDL Model
ARDL(3,3) selected Based on Akaike Information Criterion

Dependent variable is dLNY

51 observations used for estimation from 1950 to 2000

Regressor	Coefficient	Standard Error	T-Ratio[Prob]
dLNY1	-.095970	.16262	-0.59016[.558]
dLNY2	.22674	.13614	1.6655[.103]
dLNK	2.8098	.65886	4.2647[.000]
dLNK1	-3.6405	.81107	-4.4885[.000]
dLNK2	1.4008	.62414	2.2444[.030]
dC	1.4614	.52418	2.7880[.008]
ecm(-1)	-.27519	.11894	-2.3137[.025]

not significant

List of additional temporary variables created:

dLNY = LNY-LNY(-1)

dLNY1 = LNY(-1)-LNY(-2)

dLNY2 = LNY(-2)-LNY(-3)

dLNK = LNK-LNK(-1)

dLNK1 = LNK(-1)-LNK(-2)

dLNK2 = LNK(-2)-LNK(-3)

dC = C-C(-1)

ecm = LNY - .61463*LNK - 5.3106*C

R-Squared	.49886	R-Bar-Squared	.41728
S.E. of Regression	.037365	F-stat.	F(6, 44) 7.1342[.000]
Mean of Dependent Variable	.023959	S.D. of Dependent Variable	.048948
Residual Sum of Squares	.060034	Equation Log-likelihood	99.6231
Akaike Info. Criterion	91.6231	Schwarz Bayesian Criterion	83.8958
DW-statistic	1.8804		

R-Squared and R-Bar-Squared measures refer to the dependent variable dLNY and in cases where the error correction model is highly restricted, these measures could become negative.

4,

Dependent Variable: D(LNYRMFG)

Method: Least Squares

Date: 04/26/05 Time: 17:18

Sample (adjusted): 1950 2000

Included observations: 51 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
<u>C</u>	4.95E-08	0.014587	3.39E-06	1.0000
D(LNYRMFG(-1))	-0.095968	0.150418	-0.638011	0.5268
D(LNYRMFG(-2))	0.226739	0.128817	1.760161	0.0853
D(LNKRMFG)	2.809857	0.618358	4.544062	0.0000
D(LNKRMFG(-1))	-3.640514	0.801329	-4.543095	0.0000
D(LNKRMFG(-2))	1.400811	0.592546	2.364056	0.0226
<u>ECM1(-1)</u>	-0.275187	0.095832	-2.871574	0.0063
R-squared	0.498864	Mean dependent var	0.023959	
Adjusted R-squared	0.430527	S.D. dependent var	0.048948	
S.E. of regression	0.036938	Akaike info criterion	-3.632278	
Sum squared resid	0.060034	Schwarz criterion	-3.367126	
Log likelihood	99.62310	F-statistic	7.300086	
Durbin-Watson stat	1.880434	Prob(F-statistic)	0.000019	

 $dc \equiv 0$ since $c = \text{const}$

This is same
as Microfit
(see p. 4 B)
except they
run it with
 $dc \equiv 0$
(since $c = \text{const}$)

Capacity Utilization & Shocks

2/20/05

1. ln Unk1 shows sharp drops in 1973-75, 1979-82

(i) Intuitively, these are drops due to the oil shocks, not to structural relation between $\log Y^*$ and $\log K, \log V$

(ii) How should we capture this effect, or rather leave it in $\log u$ instead of shifting it to the dummies?

(iii) Adding dummies to the regressor will transform the residual into $\log u'$, i.e. into the rate of capacity utilization that would have been in the absence of (oil) shocks.

— But of course, what we need to know is the rate of capacity utilization that actually was, including oil shocks.

2. Dummies & their effects

(i) Additive dummies are shifts in the intercept of the regression

In the simulation,
have $\log Y$ drop by
equal amounts.

held as a cyclical, endless, process

6/4/08

1. Suppose $u = AR(z)$ ($u \equiv \ln U$)

$$AR(z) \begin{cases} (1-p_1L)(1-p_2L)u = \varepsilon_u \\ (1-(p_1+p_2)L + p_1p_2L^2)u = \varepsilon_u \\ u - (p_1+p_2)u_{-1} + (p_1p_2)u_{-2} = \varepsilon_u \end{cases}$$

2. Gandolfo: $u + a_1u_{-1} + a_2u_{-2} = 0$

$\rightarrow a_1 = -(p_1+p_2) \quad a_2 = p_1p_2 \quad \Delta \equiv a_1^2 - 4a_2$

Oscillatory paths require $\Delta < 0$

Stability requires $|a_2| < 1$?

(1) $\Delta = [-(p_1+p_2)]^2 - 4p_1p_2 = p_1^2 + 2p_1p_2 + p_2^2 - 4p_1p_2 = p_1^2 - 2p_1p_2 + p_2^2$

$\Delta = (p_1 - p_2)^2 \rightarrow \Delta > 0$ for any $p_1 - p_2 \neq 0$, since $(p_1 - p_2)^2 > 0$

$\rightarrow$ so oscillatory path is not possible for any $AR(z)$!
definite part of

From: <Moudud@aol.com>
To: <shaikh@newschool.edu>
Date: 10/8/2005 12:14:40 PM
Subject: Econometric sections

*Superseded
we moved to Extended Perm
Tables*

Anwar

Here it is finally. After a lot of waffling I decided to stick to the OECD Economic Outlook database from 2002 as the latest one has Japan's capacity utilization rate from 1975 whereas the earlier has it from 1970. I kept going between that and the IMF data and still haven't made up my mind. Which comparison is the most stark, do you think?

I have both OECD and IMF measures of u in the attached workfile.

Also, I was mildly concerned about France's u for the period 1975-1988 when it rises steadily although that was a crisis period. However, the mean gY for that period is 2.5% while the mean gY^* is 1.8%-hence the rise in u . That is, one can paradoxically have a rise in u in a crisis if the growth of capital accumulation is slower than output growth. Does that make sense?

Further, are we keeping Spain despite that spike? If you recall that's there in the PWT 6.1, as we checked.

Finally, I dropped Luxembourg since neither the IMF nor the OECD has a measure of u for it.

I doubt I can discuss this at length either Sunday or Monday. But I am free Tuesday morning if you want to chew the fat. Tomorrow I have to prepare for my Monday classes.

Jamee

11
4/4/05

1. From the identity $u = \frac{Y}{Y^*} = \frac{Y}{K} \left(\frac{K}{Y^*} \right) = \frac{Y}{K} v$, we get

$$1) \ln Y = \ln K + \ln u - \ln v$$

2. From the technical change function

$$2) \Delta \ln v = a_0 + b_1 \Delta \ln K + \dots$$

$$2)' v = a_0 + a_1 t + b_1 \ln K + e_v$$

we get from eq 2

$$3) \ln Y = -a_0 - a_1 t + (1-b_1) \ln K + e_u$$

where $e_u = \ln u + e_v$

LR only

2. Our problem is: how do we estimate the long run coefficients appropriate to eq 3?

3. Cointegration Approach (linear time trend in data series, for and in coint. eq. (eq 3))

I(0) if $\ln Y = I(1)$

I(0) if $\ln Y, \ln K = I(1)$ AND COINTEGRATION

OLS1: $\Delta \ln Y = \delta_1 + \delta_2 (\ln Y_{t-1} - a_0 - a_1 t + (1-b_1) \ln K_t) + \sum_{i=1}^p \lambda_i \Delta \ln Y_t + \sum_{i=1}^q \gamma_i \Delta \ln K_t + e_{1t}$

OLS2: $\Delta \ln K = \delta_2 + \delta_2 (\ln Y_{t-1} - a_0' - a_1' t + (1-b_1') \ln K_t) + \sum_{i=1}^p \lambda_i' \Delta \ln Y_t + \sum_{i=1}^q \gamma_i' \Delta \ln K_t + e_{2t}$

(I) This is valid for OLS only if

a) $\ln Y, \ln K$ are $I(1)$

b) $\ln Y$ and $\ln K$ are cointegrated

(II) Moreover it still requires us to determine appropriate lag lengths

(III) If there is a single coint. relation, then (say) $\delta_2 = 0$

LR and SR

Over Capacity Utiliz. Problem

2/22/08

1. We began from the insight that capacity (Y^*) is that part of output that is linked to the capital stock in the LR

$$1) Y^* \equiv K/v$$

$$1)' \log Y^* = \log K - \log v$$

Random (I(0)), RW = I(1), ARMA.)

2. If capacity utiliz. is I(0) in the LR, then

$$2) \log Y - \log Y^* = \log Y - (\log K - \log v) = \log u \text{ is also I(0)}$$

- (1) If $\log u \approx 0$ in the LR ($u \approx 1$), then $\log Y \approx \log Y^*$

so that we arrive at a LR relationship between Y, K

(If) $3) \log v = b_0 + b_1 t + b_2 \log K + \varepsilon_v$, then

4) $\log Y = a_0 + a_1 t + b_1 \log K + \varepsilon_Y$ in the LR ($u \approx 1$)

(11) "It is important to recognize that the existence of a long-run relationship between [two variables] y_t and x_t does not rest on whether x_t is I(1)"

[Pesaran, 1997, "The Role of EU Theory in Modelling the LR", p. 183]

- The problem is that testing the hypothesis "is a non-standard testing problem" (Pesaran 1997, pp 10, p 183)

- Pesaran, Shin & Smith, 1996, show how to do this when x_t is I(0) or I(1) [Pesaran 1997, pp 10, p 183]

Evidently, $\log Y$ ($\log Y^*$) will then have the same order of integration as $\log K$, which is what we found!

Our Capacity Utilization Measure

3/28/05

1. We want LR relation between $\ln Y$ and $\ln K$
 $\rightarrow$ implied LR coeff on $\ln K$, via an ARDL (not actual coeff)
2. Beginning from identity & specifying $\log v = f(\log K)$
gives us $\ln u$ as the residual of the identity
 $\ln u = \log Y + \log v - \ln K = \ln Y - F(\ln K)$
3. We want $\log u$ to capture both structural relations and shocks

But dummies
may not affect
the LR coefficient.

- (1) If we add dummies to the regression, which are not in the identity, then we are in effect splitting $\ln u$ into two parts
 $\rightarrow \ln u = \ln u^* + \text{dummy effects}$
where $\ln u^* =$ the capacity utilization that would have occurred in the absence of shocks

- (2) But we want the whole $\ln u$
- This is peculiarity of our problem, which is not to minimize the residual

4. So we

- (1) Do want to use structural breaks in unit roots test
To see order of integration of variables and hence LR estimation method to be used
- (2) Do not want to use structural breaks in OLS or coint.

What is relation of ARDL to this?

Given the sample procedure

(see Engles, p. 336)

$$y_t = \beta_0 + \beta_1 x_t + e_t$$

2) $e_t = a_1 e_{t-1} + \epsilon_t \rightarrow$ see if ϵ_t is white noise

are $I(0)$, this is equiv to AR(1) model $\rightarrow$ coeff β_0, β_1 are based w/o AR(1) term $\rightarrow$ ARDL
are $I(1)$, this is coint. model $\rightarrow$ coefficients β_0, β_1 are not based if y, x are $I(1)$

(- what do we do about ~~reliability~~ of coeffs since there is serial correlation ($DW \ll 2$) [see Engles p. 337 is not a problem for $I(1)$ variables])

$$\begin{cases} y = \beta_0 + \beta_1 x + e \\ e = \rho e_{-1} + \varepsilon \end{cases}$$

$$\rho y_{-1} = \rho \beta_0 + \rho \beta_1 x_{-1} + \rho e_{-1}$$

$$y - \rho y_{-1} = (1-\rho)\beta_0 + \beta_1 x - \rho \beta_1 x_{-1} + \underbrace{e - \rho e_{-1}}_{\varepsilon}$$

$$\rightarrow y = (1-\rho)\beta_0 + \rho y_{-1} + (\beta_1 x - \rho \beta_1 x_{-1}) + (\varepsilon)$$

$$\rho = -0.7076$$

↑
AR(1) se

$\log K$, $\log I$ and $\log Y$ are all $I(1)$ (if) there is no tech. change

1. An OLS requires that each variable or composite variable be $I(0)$

(1) This is required for constant mean + var, etc., in OLS

2. Integration (Enders 2004)

(1) Trend stationary

$$y_t = [y_0 + a_0 t] + A(L) \varepsilon_t, \quad A(L) = 1 - L - L^2 - \dots = \varepsilon_t - \varepsilon_{t-1} - \varepsilon_{t-2} - \dots$$

$$\Delta y_t = a_0 + [A(L) \Delta \varepsilon_t]$$

(158, Enders 2004)

[e.g., $\Delta y_t = a_0 + \Delta \varepsilon_t$, $A(L) = 1$]

(1) (11) Stochastic Trend

$$y_t = [y_0 + a_0 t] + \sum_{k=1}^t \varepsilon_k \quad (159)$$

$$g_K = \Delta \log K = \frac{\Delta K}{K} + \varepsilon_t$$

(160)

(Random Walk with Drift)

$$\log K = [\log K_0 + \beta(1) \cdot t] + \sum_{k=1}^t \varepsilon_k \quad (161)$$

$$\Delta y_t = a_0 + \varepsilon_t$$

$$y_t = y_{t-1} + a_0 + \varepsilon_t \quad (161)$$

$$I_t = g_K K_{t-1} = \left[\frac{\Delta K}{K} + \varepsilon_t \right] K_{t-1}$$

(160) (11) Random Walk

$$y_t = [y_0] + \sum_{k=1}^t \varepsilon_k \quad (159)$$

$$\log I = \log(g_K + \varepsilon_t) + \log K_{t-1}$$

$$\Delta y_t = \varepsilon_t \quad (159)$$

$$y_t = y_{t-1} + \varepsilon_t$$

$\therefore \log I \approx \log I_{t-1} + \varepsilon_t$

But $y = I/K \Rightarrow \log y \approx \log I$

$$I_t = \Delta y_t, \quad g_K = \frac{\Delta K}{K} + \varepsilon_t$$

$$y_t = y_{t-1} + \varepsilon_t$$

$$g_K = \frac{\Delta K}{K} + \varepsilon_t \Rightarrow \varepsilon_t = g_K - \frac{\Delta K}{K}$$

(160)

(11) Random Walk with noise

$$y_t = y_0 + \sum_{k=1}^t \varepsilon_k + \eta_t \quad (162)$$

$$\Delta y_t = \varepsilon_t + \Delta \eta_t \quad (163)$$

$\Delta \log K = (\bar{S} \cdot R_t) \Delta K_t + \varepsilon_t$ if K is constant, in general,

(161) (11) Stochastic Trend with noise

$$y_t = [y_0 + a_0 t] + \sum_{k=1}^t \varepsilon_k + \eta_t \quad (164)$$

$$\Delta y_t = a_0 + \varepsilon_t + \Delta \eta_t$$

$$y_t = y_{t-1} + a_0 + \varepsilon_t + \eta_t$$

(161) Stochastic and Det. Trend with noise $y_t = [y_0 + a_0 t] + \sum_{k=1}^t \varepsilon_k + A(L) \eta_t \quad (164)$

$$\Delta y_t = a_0 + \varepsilon_t + A(L) \Delta \eta_t$$

WORKFILE KEY

All equations with "star0" as an attachment are the conditional ECMs and their respective ADLs. All these equations show the existence of long-run relationships between $\ln y$ and $\ln k$ since the t - and F -statistics are significant in terms of Pesaran's (2001) critical values. Remember to check the conditional ECM to ensure that it is free of serial correlation.

All equations with "star" only as the attachment are the best "pruned-down" ADL equations and have no serial correlation, no heteroscedasticity, no non-normal errors, and no functional form misspecification.

All equations with "a", "b", "c" etc at the end correspond to different model specifications all of which produced reasonably good results. No letter is attached if only one estimation was made.

Example:

$eqecmusstar0a$ is the conditional ECM for the US and the corresponding conditional ADL is $eqadlstar0a$. The best ADL obtained from this is $eqadlustara$. The corresponding capacity is $yusstara$ and capacity utilization rate $uuusa$. All equations of the type " $eqecmrepadlusstar$ " are the ECM representations of the best ADLs and allow us to obtain the t -statistic of the ECM coefficient.

Dummies of the form $d80to82$ take the value of 1 between the years specified and 0 elsewhere.

Finally, ignore any equations with "old" or "new" as attachments.

R = output/capital ratio

R^* = capacity/capital ratio

$uimf$ = IMF estimates of capacity utilization rates from World Economic Outlook online database 2002

$uoecd72$ = capacity utilization rate from OECD Economic Outlook 72

$uoecd77$ = capacity utilization rate from OECD Economic Outlook 77

The measure of u from the OECD Economic Outlook 72 or 77 is derived from the output gap which is defined as (OECD website):

"Definition: The output gap is measured as the percentage difference between actual GDP in constant prices, and estimated potential GDP. The latter is estimated using a production function approach for all countries except Portugal, taking into account the capital stock, changes in labour supply, factor productivity and underlying non-accelerating wage rates of unemployment or the NAWRU for each Member country. Potential output for Portugal is calculated using a Hodrick-Prescott filter of actual output. It should be stressed that the estimated levels of potential output are subject to significant margins of error. See Giorno et al. (1995) and Turner et al. (1996)."

COUNTRY CODES

aus = Australia, aut = Austria, bel = Belgium, can = Canada, che = Switzerland, dnk = Denmark, esp = Spain, fra = France, fin = Finland, gbr = Great Britain, grc = Greece, ita = Italy, irl = Ireland, jpn = Japan, lux = Luxembourg, nld = Netherlands, nor = Norway, prt = Portugal, swe = Sweden, and us = United States.

GOODNESS-OF-FIT CRITERIA

The null hypotheses are:

1) H_0 : no serial correlation

The test for serial correlation is based on the LM and F statistics which have the same distribution asymptotically. Harvey (1981, p. 277) also refers to the F stat as the 'modified LM' statistic. In the MICROFIT manual (p. 86) Pesaran and Pesaran state that "on the basis of Monte Carlo results, Kiviet (1986) has shown that in small samples the F -version is generally preferable to the LM version". Pesaran and Pesaran report (MICROFIT manual, p. 86) that the lag p for the serial correlation LM test should be 1 for annual data (MICROFIT does this automatically).

2) H_0 : no heteroscedasticity

3) H_0 : no non-normal residuals (Jacque-Bera statistic)

4) H_0 : no functional form misspecification (Ramsey RESET statistic)

It is not a major problem if the model fails the RESET test since Pesaran's (2001) wage model fails this test. One reason why a model may fail this test is because of the presence of possible nonlinearities.

GOOD RESULTS

1. TIME & Classical/Hammond
"CH" not "AM"

2. Agreements/Disagreements

(i) SR : DD, etc $\rightarrow$ general "agreement" $\leftarrow \begin{matrix} \text{GDP} \uparrow \rightarrow P \uparrow \\ W \uparrow \rightarrow L \uparrow \end{matrix}$
[They see SR as "entirely" PK]

(ii) LR

a) KK investment : $s \uparrow \rightarrow$ paradox of thrift
 $s' = s + s_n(u - v_n)$ and $u_n \rightarrow$ trend

b) Mankiw

1) $s' = s + s_n v = s + s_n u + s_n v_n$ (Krug, Staffman)
 $\rightarrow$ but this led to critique that v_n was not
"effective" unless $u \rightarrow u_n$ in LR, which
needs to be shown

2) Dornbusch/Krug

If $u > u_n$, Central Bank raises i , so

3) Shaikh/Morduch